

We consider the unconstrained optimization problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x), \quad (1)$$

where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuously differentiable nonconvex function, and $x \in \mathbb{R}^n$ is the optimization variable. We focus on settings where objective function evaluations, especially the function values, are contaminated by bounded and non-diminishing numerical noise [10]. Specifically, only inexact function values $\bar{f}(x)$ are available for optimization. Such inaccuracies are unavoidable in many realistic scenarios, including finite-precision arithmetic, simulation-based evaluations, and stochastic approximations [1, 6, 9].

In the absence of noise, quasi-Newton and regularized quasi-Newton methods can solve the large-scale problem (1), and their theoretical guarantees and practical efficiency have been well established. For their success, the accuracy of function values is crucial. Accurate function values are used to determine whether a step is accepted or rejected, and to adjust the step size or regularization parameter [4, 8, 11–13]. When function values are contaminated by numerical noise, these mechanisms become unreliable, and the algorithms may lose their theoretical guarantees, behave erratically, or terminate prematurely with unreliable solutions.

In the presence of noise, there are mainly two approaches to address function-value errors. The first is to use noise-tolerant methods that account for bounded errors, such as the noise-tolerant quasi-Newton (NTQN) method [9]. The second is to avoid function values entirely, as in Objective-Function-Free Optimization (OFFO) [2, 3]. However, the NTQN convergence analysis covers only strongly convex objectives, and totally discarding function values can limit practical efficiency when reliable function values are available.

These observations naturally motivate us to design noise-tolerant quasi-Newton methods that use function values when they are reliable and use noise-aware strategies when they are not. A natural idea is to switch between quasi-Newton steps guided by function values and OFFO-inspired regularized steps. Still, overlooking function-value errors can compromise convergence guarantees, whereas overly conservative switching can slow practical convergence. The key difficulty is therefore to determine when to switch while maintaining both convergence guarantees and practical efficiency.

In this work, we propose a noise-tolerant regularized quasi-Newton method. Our approach combines Armijo-type line search with quadratic regularization [4, 12] and an OFFO-inspired strategy [2, 3]. When a sufficient decrease in function value is observed, the method behaves similarly to standard quasi-Newton methods with line search, preserving its practical efficiency. Otherwise, an OFFO-inspired cumulative-gradient scale controls the regularization and supports convergence under function-value errors.

The key methodological advance is the carefully designed, error-model-based switch between unregularized quasi-Newton steps and OFFO-inspired regularized steps. A relaxed Armijo condition alone may accept steps without sufficient decrease in the true objective, so its decrease estimates need not establish convergence to arbitrarily small first-order tolerances. Our switch permits unregularized quasi-Newton

Table 1: Theoretical comparison of the proposed method, related Newton-type, quasi-Newton, and OFFO methods. The entries summarize theoretical guarantees, rather than the settings considered in numerical experiments. A checkmark (✓) denotes that the corresponding property holds or is covered by the cited theory, and “–” denotes that it does not hold or is not established by that theory. The “Noisy f ,” “Noisy ∇f ,” and “Nonconvex” columns indicate whether the corresponding convergence theory allows function-value errors, allows gradient errors, and covers nonconvex objectives, respectively. For OFFO, function-value errors have no effect because function values are not evaluated.

Method	Uses f	Noisy f	Noisy ∇f	Nonconvex	$\mathcal{O}(\varepsilon^{-2})$ guarantee
L-BFGS [5, 8, 13]	✓	–	–	✓	–
Regularized Newton [11]	✓	–	–	✓	✓
Regularized L-BFGS [4]	✓	–	–	✓	–
NTQN [9, 14]	✓	✓	✓	–	–
OFFO [3]	–	✓	–	✓	✓
Proposed (??)	✓	✓	–	✓	✓

steps only when the observed progress is sufficient to account for function-value errors. This allows the method to exploit reliable function values while retaining convergence guarantees without applying OFFO-inspired regularization throughout. The resulting method achieves an $\mathcal{O}(\varepsilon^{-2})$ iteration bound, preserving the optimal worst-case order for smooth nonconvex optimization under the problem setting of persistent function-value errors and exact gradients.

We also conducted extensive numerical experiments on the CUTEst benchmark collection to compare the analyzed method and a heuristic variant with existing methods. We evaluated their performance under function-value noise within the theoretical problem setting, and under joint function-and-gradient noise and reduced-precision evaluations as complementary empirical tests beyond this setting. We also assessed sensitivity to error-bound misspecification. Both proposed variants attain higher solved percentages than the existing quasi-Newton baselines under artificial noise and at reduced precision. A fixed-iteration timing experiment shows lower overhead than the regularized L-BFGS baselines, and a separate time-to-solution comparison shows favorable elapsed-time performance under reduced precision.

0.1 Comparison With Related Work

We next discuss the methods in Table 1 in order, focusing on their assumptions and convergence guarantees.

The first three methods in the table are standard quasi-Newton and regularized Newton-type methods. The Broyden–Fletcher–Goldfarb–Shanno (BFGS) method and its Limited-memory variant (L-BFGS) are among the most efficient and widely used algorithms to solve the problem (1) [5, 7]. They generally employ line search procedures that enforce the Wolfe conditions to guarantee sufficient descent and the positive definiteness of the Hessian approximation. Regularized Newton methods attain

an $\mathcal{O}(\varepsilon^{-2})$ iteration bound under their stated assumptions and with accurate function values [11, Theorem 4.2]. The cited analysis of regularized L-BFGS establishes only asymptotic global convergence to first-order stationarity [4, Theorem 2]. As discussed above, their theoretical guarantees rely on accurate function values, and do not cover function-value errors. Their practical performance may degrade in noisy environments.

NTQN addresses bounded errors in function and gradient evaluations through a noise-tolerant line search and a lengthening procedure [9]. The NTQN analysis allows bounded gradient errors and establishes convergence to a noise-dependent neighborhood for strongly convex objectives [9, Theorem 3.5]. Its practical line search adds twice the absolute function-error bound to the Armijo right-hand side after an unsuccessful initial trial. Our relaxation instead follows a hybrid absolute-relative error model and is recomputed at each trial point.

OFFO methods obtain convergence guarantees without evaluating function values [2, 3]. The Adagrad-like class analyzed by Gratton et al. [3] attains an $\mathcal{O}(\varepsilon^{-2})$ iteration bound for smooth nonconvex objectives, but does not use function-value information to guide the steps.

Lastly, our method uses inexact function values to permit unregularized quasi-Newton steps when progress is certified and retains this worst-case order under persistent function-value errors.

0.2 Organization of the Paper

References

1. Caffisch RE (1998) Monte Carlo and quasi-Monte Carlo methods. *Acta Numerica* 7:1–49, DOI 10.1017/S0962492900002804
2. Gratton S, Toint PL (2023) OFFO minimization algorithms for second-order optimality and their complexity. *Computational Optimization and Applications* 84(2):573–607, DOI 10.1007/s10589-022-00435-2
3. Gratton S, Jerad S, Toint PhL (2026) Complexity of a class of first-order objective-function-free optimization algorithms. *Optimization Methods and Software* 41(2):478–508, DOI 10.1080/10556788.2023.2296431, <https://doi.org/10.1080/10556788.2023.2296431>
4. Kanzow C, Steck D (2023) Regularization of limited memory quasi-Newton methods for large-scale nonconvex minimization. *Mathematical Programming Computation* 15(3):417–444, DOI 10.1007/s12532-023-00238-4
5. Liu DC, Nocedal J (1989) On the limited memory BFGS method for large scale optimization. *Mathematical Programming* 45(1):503–528, DOI 10.1007/BF01589116
6. Moré JJ, Wild SM (2011) Estimating Computational Noise. *SIAM Journal on Scientific Computing* 33(3):1292–1314, DOI 10.1137/100786125
7. Nocedal J, Wright SJ (1999) *Numerical Optimization*. Springer, DOI 10.1007/978-0-387-40065-5

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8. Okazaki N (2007) libLBFGS: A library of limited-memory broyden-fletcher-goldfarb-shanno (l-BFGS). URL <https://github.com/chokkan/liblbfgs>
 9. Shi HJM, Xie Y, Byrd R, Nocedal J (2022) A Noise-Tolerant Quasi-Newton Algorithm for Unconstrained Optimization. *SIAM Journal on Optimization* 32(1):29–55, DOI 10.1137/20M1373190
 10. Sun S, Nocedal J (2023) A trust region method for noisy unconstrained optimization. *Mathematical Programming* 202(1):445–472, DOI 10.1007/s10107-023-01941-9
 11. Ueda K, Yamashita N (2010) Convergence Properties of the Regularized Newton Method for the Unconstrained Nonconvex Optimization. *Applied Mathematics and Optimization* 62(1):27–46, DOI 10.1007/s00245-009-9094-9
 12. Ueda K, Yamashita N (2014) A regularized Newton method without line search for unconstrained optimization. *Computational Optimization and Applications* 59(1):321–351, DOI 10.1007/s10589-014-9656-x
 13. Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, Burovski E, Peterson P, Weckesser W, Bright J, van der Walt SJ, Brett M, Wilson J, Millman KJ, Mayorov N, Nelson ARJ, Jones E, Kern R, Larson E, Carey CJ, Polat I, Feng Y, Moore EW, VanderPlas J, Laxalde D, Perktold J, Cimrman R, Henriksen I, Quintero EA, Harris CR, Archibald AM, Ribeiro AH, Pedregosa F, van Mulbregt P, SciPy 10 Contributors (2020) SciPy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods* 17:261–272, DOI 10.1038/s41592-019-0686-2
 14. Xie Y, Byrd RH, Nocedal J (2020) Analysis of the BFGS Method with Errors. *SIAM Journal on Optimization* 30(1):182–209, DOI 10.1137/19M1240794